

RAJENDRA VIDYALAYA

UNIT TEST — CHEMISTRY

Class VIII

Maximum Marks: 20

Name: _____ Roll No.: _____ Date: _____

Instructions: Answer all questions. Draw neat, labelled diagrams where required.

1. Fill in the blanks. ($5 \times 1 = 5$ marks)

- (a) _____ are positively charged subatomic particles present in the nucleus of an atom.
- (b) The _____ of an element is equal to the number of protons in its nucleus.
- (c) _____ are electrically neutral subatomic particles present in the nucleus.
- (d) Protons and neutrons together are known as _____.
- (e) A negatively charged ion is called a/an _____.

2. Write two main features of Rutherford's atomic model. (2 marks)

3. An atom has atomic number 4 and 5 neutrons. Find its mass number, name the element and write its symbol in standard notation. (3 marks)

4. Define isotopes. Give three isotopes of hydrogen using the correct notation. (3 marks)

5. For each of the following atoms: ($2 \times 3\frac{1}{2} = 7$ marks)

- (i) Aluminium-27 (atomic number 13)
- (ii) Chlorine-35 (atomic number 17)
- (a) Draw the atomic (Bohr) diagram.
- (b) State the mass number (A).
- (c) State the atomic number (Z).
- (d) Find the number of neutrons.
- (e) Find the number of protons.
- (f) Write the electronic configuration.